

(1001CJA102120043)

Test Pattern



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2020 - 2021)

JEE(Advanced)
PART TEST
09-01-2021

JEE(Main + Advanced) : ENTHUSIAST COURSE (SCORE-I)

ANSWER KEY

PAPER-1

PART-1 : PHYSICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A,B,C	A,C,D	A,B,C	B,C	A,B,C	B	B	D	B	C
SECTION-II	Q.	1	2	3	4	5	6	7	8		
	A.	0.59	0.80	8.00	46.00	20.00	4.00	12.00	90.40 to 90.55		

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	C	B	A,C	B,C,D	D	A,B,C	B	A	B	B
SECTION-II	Q.	1	2	3	4	5	6	7	8		
	A.	9.00	5.00	4.00	7.00	2.50 to 2.60	3.60	2.00	2.00		

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A,D	A,C	A,B,C	A,C	A,C,D	A,C	C	A	D	A
SECTION-II	Q.	1	2	3	4	5	6	7	8		
	A.	2007.00	2.00	3.00	8.00	1024.00	0.00	7.00	5.00		

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CLASSROOM CONTACT PROGRAMME

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JEE(Main)
PART TEST
09-01-2021

JEE(Main + Advanced) : ENTHUSIAST COURSE (SCORE-I)

ANSWER KEY

PAPER-2

PART-1 : PHYSICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	C	A	A	B	A	D	A	A	B	D
SECTION-II	Q.	11	12	13	14	15	16	17	18	19	20
	A.	A	D	A	C	B	C	C	C	A	A
SECTION-II	Q.	1	2	3	4	5	6	7	8	9	10
	A.	3.00	51.24	6.00	1.00	0.50	3.09	42.00	6.00	2.28	0.16

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	D	B	C	A	C	C	C	C	C	B
SECTION-II	Q.	11	12	13	14	15	16	17	18	19	20
	A.	D	B	D	B	C	B	A	C	B	D
SECTION-II	Q.	1	2	3	4	5	6	7	8	9	10
	A.	130.00	11.00	11.00	5.00	2.00	10.00	5.00	8.00	4.00	8.00

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A	B	D	C	C	D	B	C	B	A
SECTION-II	Q.	11	12	13	14	15	16	17	18	19	20
	A.	C	D	C	B	C	B	C	A	A	B
SECTION-II	Q.	1	2	3	4	5	6	7	8	9	10
	A.	25.00	7.00	1.00	3.00	20.00	0.25	0.25	0.85 or 0.86	0.25	1.25

JEE(Main + Advanced) : ENTHUSIAST COURSE (SCORE-I)**PAPER-1****PART-1 : PHYSICS****SOLUTION****SECTION-I**1. **Ans. (A,B,C)**

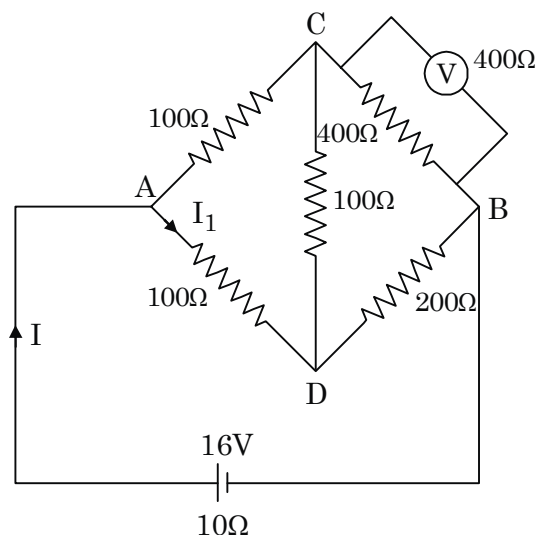
Sol. $v_d = \frac{eE}{m} \tau$

$$R \propto \rho \propto \frac{1}{\tau}$$

as temperature increase relaxation time decrease.

2. **Ans. (A,C,D)**

Sol. In the balanced Wheatstone bridge, 100Ω resistor between CD is null and develops zero power.



$$I = \frac{16}{\frac{300}{2} + 10} = 0.1 \text{ A}$$

$$I = 0.05 \text{ A}$$

$$\text{Voltmeter reading} = 0.025 \times 400 = 10 \text{ V}$$

3. **Ans. (A,B,C)**

Sol. In series error is $\Delta R_1 + \Delta R_2$

In parallel :

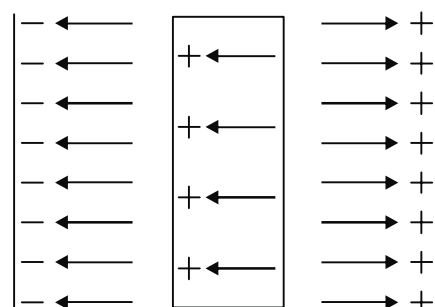
$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2}$$

$$\frac{\Delta R_{eq}}{R_{eq}^2} = \frac{\Delta R_1}{R_1^2} + \frac{\Delta R_2}{R_2^2}$$

$$\Delta R_{eq} = \frac{1}{15} \Omega = 0.06 \Omega \approx 0.1 \Omega$$

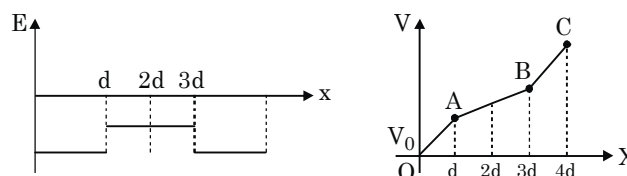
4. **Ans. (B,C)**

Sol. The magnitude and direction of electric field in different region are shown. The direction of the electric field remains the same, hence option (B) is correct.

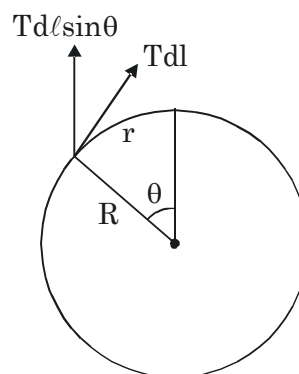


Similarly, electric lines always flow from higher to lower potential, therefore electric potential increases continuously as we move from $X = 0$ to $X = 4d$ therefore, option (c) is correct.

The variation of electric field (E) and potential (V) with x will follows

5. **Ans. (A,B,C)**

Sol. The situation of drop is clearly shown in figure.



The net vertical force in upward direction due to surface tension is,

$$F = \int T d\ell \sin \theta = T \sin \theta \times \int d\ell = T \sin \theta \times 2\pi r$$

$$= T \times \frac{r}{R} \times 2\pi r = \frac{2\pi r^2 T}{R}$$

The drop will detach from dropper when $F < mg$. For limiting case $F = mg$

$$\Rightarrow \frac{2\pi r^2 T}{R} = \frac{4\pi R^3}{3} \times \rho g \Rightarrow R^4 = \frac{3r^2 T}{2\rho g}$$

$$\Rightarrow R = \left(\frac{3r^2 T}{2\rho g} \right)^{1/4} = \left[\frac{3 \times 25 \times 10^{-8} \times 0.11}{2 \times 10^3 \times 10} \right]^{1/4}$$

$$= 1.4 \times 10^{-3} \text{ m}$$

The surface energy of the drop is, after it detached from dropper would be given by

$$U = 4\pi R^2 \times T = 4\pi \times (1.4 \times 10^{-3})^2 \times 0.11$$

$$= 4\pi \times 1.96 \times 10^{-6} \times 0.11 = 2.7 \times 10^{-6} \text{ J}$$

6. Ans. (B)

Sol. Pressure differential across capillary is

$$p_i - p_0 = \rho gh$$

$$= 1000 \times 10 \times 0.05 = 500 \text{ Nm}^{-2}$$

Viscosity of water is

$$\eta = \left(0.8 \text{ cp} \times \frac{10^{-3}}{\text{cp}} \text{ Kgm}^{-1} \text{ s}^{-1} \right)$$

$$= 8 \times 10^{-4} \text{ kgm}^{-1} \text{ s}^{-1}$$

Flow rate using Poiseuille's law is

$$j = \frac{\pi r^4 (p_i - p_0)}{8\eta L}$$

$$= \frac{\pi \times 1 \times 10^{-4} \times 5000}{8 \times 8 \times 10^{-4} \times 0.2}$$

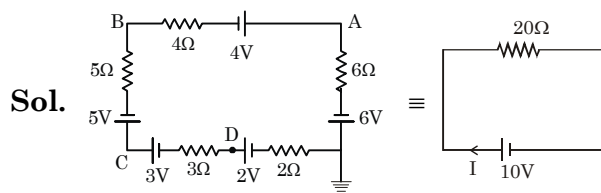
$$= \frac{5\pi}{128} \times 10^{-5} \text{ m}^3 \text{ s}^{-1} \approx 1 \text{ mL s}^{-1}$$

7. Ans. (B)

Sol. The change in apparent weight is equal to change in buoyancy force hence there is no change in the position of block wrt water surface.

As air pressure increases, according to Pascal's law this change in pressure is transmitted to each part of water and hence buoyancy force remains same and hence fraction of block above the water remains same.

8. Ans. (D)



$$\Rightarrow I = \frac{1}{2} \text{ A}$$

$$V_A - 6 \times 1/2 + 6 = 0 \Rightarrow V_A = -3\text{V}$$

$$V_B - 4 \times 1/2 + 4 = V_A = -3 \Rightarrow V_B = -5\text{V}$$

$$0 - 2 \times 1/2 + 2 - 3 \times 1/2 + 3 = V_C = 2.5\text{V}$$

$$0 - 2 \times 1/2 + 2 = V_D \Rightarrow V_D = 1\text{V}$$

9. Ans. (B)

Sol. In Case of charging of capacitor,

$$q = q_0 \left(1 - e^{-\frac{t}{\tau}} \right), \tau = RC$$

$$I = \frac{dq}{dt} = I_0 e^{-\frac{t}{\tau}}$$

$$= I_{\max} \text{ at } t = 0 \rightarrow (P) \rightarrow 2$$

$$\text{Also, } q = \frac{q_0}{2} \text{ at } t = \tau \ln 2 = RC \ln 2$$

$$\Rightarrow (R) \rightarrow (1, 3, d)$$

$$U = \frac{q^2}{2C}, P = \frac{dU}{dt} = \frac{qI}{2C} = \frac{q_0 \left(1 - e^{-\frac{t}{\tau}} \right) I_0 e^{-\frac{t}{\tau}}}{2C}$$

$$\text{for } P_{\max}, \frac{dP}{dt} = 0 \Rightarrow t = \tau \ln 2$$

$$\text{Also, } \frac{1}{2} \times \frac{q_0^2}{2C} = \frac{q_0^2}{2C} \left(1 - e^{-\frac{t}{\tau}} \right)^2$$

$$\Rightarrow t = 1.23\tau$$

10. Ans. (C)

Sol. $E_2 = \frac{E_1}{L} \times \ell$ where L = length of AB

(A) If E_1 is increased, ℓ should be decreased keeping E_2 and L constant.

(B) If R is increased, current through potentiometer will decrease. Hence V_{AJ} will decrease. But $V_{AJ} = E_2$ means length should be increased.

(C) If E_2 is increased, ℓ should be increased.

SECTION-II

1. **Ans. 0.59**

Sol. MSD = 0.1 cm

$$\text{VSD} = 9/10 \text{ MSD} = 0.9 \times 0.1 = 0.09 \text{ cm}$$

$$\text{Least count} = \text{MSD} - \text{VSD}$$

$$= 0.1 - 0.09 = 0.01 \text{ cm}$$

Total reading = Main Scale Reading + Vernier coincidence $\times$ Least count

$$1\text{st reading} = 0.5 + 8 \times 0.01 = 0.58 \text{ cm}$$

$$2\text{nd reading} = 0.5 + 4 \times 0.1 = 0.54 \text{ cm}$$

$$3\text{rd reading} = 0.5 + 6 \times 0.01 = 0.56 \text{ cm}$$

$$\text{Average reading} = (0.58 + 0.54 + 0.56)/3 = 0.56 \text{ cm}$$

$$\text{Corrected reading} = \text{reading} - \text{zero error} = 0.56 - (-0.03) = 0.56 + 0.03 = 0.59 \text{ cm}$$

2. **Ans. 0.80**

Sol. Least count

$$= \frac{\text{pitch}}{\text{no. of division on circular scale}}$$

$$= \frac{0.5 \text{ mm}}{50} = 0.01 \text{ mm}$$

$$\text{Negative zero error} = -5 \times \text{LC} = -5 \times 0.01 = -0.05 \text{ mm}$$

Measured value = main scale reading + screw gauge reading - zero error

$$= 0.5 \text{ mm} + 25 \times 0.01 \text{ mm} - (-0.05 \text{ mm})$$

$$= 0.75 \text{ mm} + 0.05 \text{ mm} = \mathbf{0.80 \text{ mm}}$$

3. **Ans. 8.00**

Sol. The electrostatic field energy of the capacitor without the dielectric slab is

$$U_1 = \frac{1}{2} CV^2$$

$$= \frac{1}{2} \times (100 \mu\text{F}) \times (200\text{V})^2 = 2 \mu\text{J}$$

and that after the slab is inserted is

$$U_2 = \frac{1}{2} \times (500 \mu\text{F}) \times (200\text{V})^2 = 10 \mu\text{J}$$

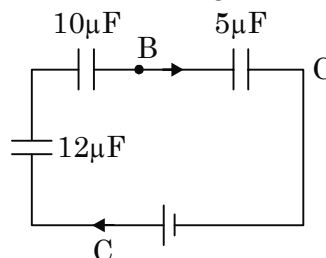
Thus, the energy is increased by 8 μJ .

4. **Ans. 46.00**

Sol. The given circuit is equivalent to as shown

$$E - \frac{q}{12} - \frac{q}{10} - \frac{q}{5} = 0$$

$$\text{Solving } q = \frac{60E}{23}$$



$$V_A - V_B = \frac{q}{12} + \frac{q}{10} = q \times \frac{22}{120} = \frac{60E}{23} \times \frac{22}{120}$$

$$\frac{11E}{23} \Rightarrow E = 46\text{V}$$

$$V_B - V_C = \frac{q}{5} = \frac{60t}{23 \times 5} = \frac{12t}{23}$$

5. **Ans. 20.00**

Sol. Loss in pressure energy = Gain in kinetic energy

$$\Delta p = \frac{1}{2} \rho v^2$$

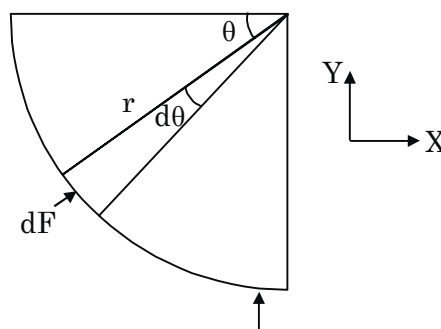
$$v = \sqrt{2 \frac{\Delta p}{\rho}}$$

$$v = \sqrt{\frac{2 \times (3-1) \text{ atm}}{10^3}}$$

$$v = \sqrt{\left\{ \frac{2 \times 2 \times 10^6}{10^3} \right\}} = 20 \text{ ms}^{-1}$$

6. **Ans. 4.00**

Sol. The force to be exerted on cylinder to keep it in equilibrium is equal and opposite of the force applied by liquid on cylinder.



Two dimensions view of the part of cylinder which is in contact with liquid

Now, we are calculating the force experienced by liquid on cylinder. For the shown element,

$$p = \rho g \times r \sin \theta$$

$$dF = p da = \rho g r \sin \theta \times [rd\theta \times \ell]$$

$$dF = \rho g \ell r^2 \sin \theta d\theta$$

$$d\vec{F} = dF \cos \theta \hat{i} + dF \sin \theta \hat{j}$$

$$\begin{aligned} \vec{F} &= \int_0^{\pi/2} dF \cos \theta \hat{i} + \int_0^{\pi/2} dF \sin \theta \hat{j} \\ &= \frac{\rho g \ell r^2}{2} \left[\hat{i} + \frac{\pi}{2} \hat{j} \right] \end{aligned}$$

So required, force $\vec{F}_1 = -\vec{F}$

$$|\vec{F}_1| = \frac{\rho g \ell r^2}{2} \sqrt{1 + \frac{\pi^2}{4}}$$

7. Ans. 12.00

Sol. Potential difference across AB = 5V

Potential difference across

$$40 \text{ cm} = \frac{5}{100} \times 40 = 2V$$

Potential difference 20 cm of wire CD = 2V

$\therefore$ Potential difference across

$$CD = \frac{2}{20} \times 80 = 8V$$

Potential difference 2Ω resistor

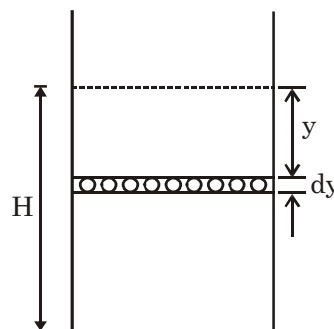
$$= 2 \times 2 = 4V$$

$\therefore$ Emf of cell = 12 volt

8. Ans. 90.40 to 90.55

Sol. Speed of efflux at a depth y below the water surface is $v = \sqrt{2gy}$

Consider a strip of height dy as shown.



Cumulative area of holes in this strip is $A = \eta (2\pi r dy)$

Rate (volume per unit time) at which the liquid flows out through the strip is

$$= vA = 2\pi \eta r \sqrt{2g} \sqrt{y} dy$$

Total rate of outflow

$$= 2\pi \eta r \sqrt{2g} \int_0^H \sqrt{y} dy$$

$$= 2\pi \eta r \sqrt{2g} \frac{2}{3} \left[y^{3/2} \right]_0^H = \frac{4}{3} \pi \eta r \sqrt{2g} H^{1/2}$$

$\therefore$ For level to remain static

$$Q = \frac{4}{3} \pi \eta r \sqrt{2g} H^{1/2} = 90.51$$

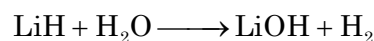
PART-2 : CHEMISTRY

SOLUTION

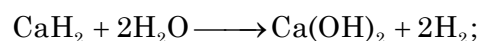
SECTION-I

1. Ans.(C)

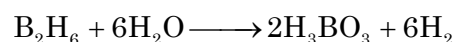
Sol. $P = \text{LiH}$, $Q = \text{CaH}_2$ and $R = \text{B}_2\text{H}_6$



$$w_{\text{H}_2} = \frac{2 \times a}{9} = \frac{2a}{9}$$



$$w_{\text{H}_2} = \frac{4 \times a}{42} = \frac{2a}{21}$$



$$w_{\text{H}_2} = \frac{a}{28} \times 6 \times 2 = \frac{2a}{4.66}$$

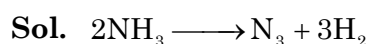
2. Ans. (B)

Sol. Two possible stereoisomers of the given compound are—



In the above shown stereoisomers, I is more polar since, the two dipole vectors $\text{C}-\text{F}$ and $\text{C}-\text{Cl}$ are at an angle of 60° and the magnitude of resultant dipole moment will be greater than either of individual bond moment. On the other hand the two dipole vectors, $\text{C}-\text{F}$ and $\text{C}-\text{Cl}$ vectors at 180° angle in II and they are in opposite direction, cancelling part of effect of one another, magnitude of resultant dipole vector will be smaller than either of the individual bond moment.

3. Ans. (A,C)



$$\begin{array}{ccc} 100 & 80 & 80 \end{array}$$

$$K_p = \frac{80 \times (80)^3}{100}$$



$$\begin{array}{ccc} 100 & 80 & 80 \end{array} \quad x = 16$$

$$\begin{array}{ccc} 100 & 80 - p & 80 \end{array}$$

$$\begin{array}{ccc} 100 - 2x & 80 - p + x & 80 + 3x \end{array}$$

$$\begin{array}{ccc} 68 & 96 - p & 128 \end{array}$$

$$K_p = \frac{(96 - p) \times (128)^3}{(68)^2} = \frac{(80)^4}{(100)^2}$$

$$(96 - p) = 9 = \text{Partial pressure of } \text{N}_2.$$

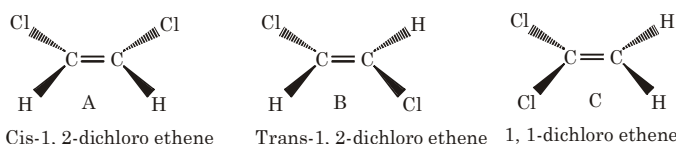
4. Ans. (B,C,D)

Sol. In isoelectronic species with increase of Z, size decreases.

All are correct.

5. Ans. (D)

Sol. Since, the molecule has one degree of unsaturation, and contains only two carbons, possibility of cyclic structure is ruled out and it must contain a carbon-carbon double bond. Three possible structures are :



6. Ans. (A,B,C)

Sol. $S = A \cdot e^{-\Delta H/RT}$

$$\Rightarrow \ln S = \ln A - \frac{\Delta H}{RT}$$

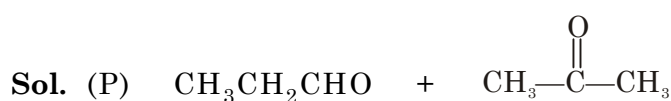
Positive slope represents that $\Delta H = \text{Negative}$.

Dissolution of CaO is exothermic.

7. Ans. (B)

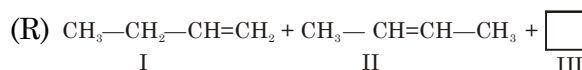
Sol. Theory based.

8. Ans. (A)



Functional  Ring-chain

(Q) $\text{CH}_3\text{CHCl}_2 + \text{CH}_2\text{ClCH}_2\text{Cl}$; Positional



I + II = Positional

I + III or II + III = Ring – chain isomers.



I + II = Functional; I + III = Positional

9. Ans. (B)

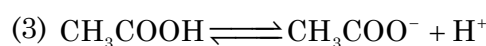
Sol. (1) $[\text{OH}^-]$ to just start precipitation

$$= \sqrt{\frac{K_{sp}}{[\text{Mg}^{2+}]}} = \sqrt{\frac{2 \times 10^{-6}}{2 \times 10^{-3}}} = 10^{-1.5}$$

$$\therefore \text{pOH} = 1.5 \Rightarrow \text{pH} = 12.5$$

$$(2) [\text{OH}^-]_{\max} = \left(\frac{K_{sp}}{[\text{Al}^{3+}]} \right)^{1/3} = \left(\frac{10^{-28}}{0.1} \right)^{1/3} = 10^{-9} \text{ M}$$

$$\therefore \text{p}_{\min}^{\text{OH}} = 9 \Rightarrow \text{p}_{\max}^{\text{H}} = 5.0$$



$$\left(0.1 - 0.1 \times \frac{10}{11} \right) \text{ M} \quad 0.1 \times \frac{10}{11} \text{ M}$$

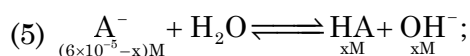
$$= \frac{0.1}{11} \text{ M} \quad = \frac{1}{11} \text{ M}$$

$$\text{Now, } K_a = 10^{-5} = \frac{\frac{1}{11} \times [\text{H}^+]}{\frac{0.1}{11}} \Rightarrow [\text{H}^+] = 10^{-6} \text{ M}$$

$$\therefore \text{pH} = 6.0$$

$$(4) [\text{H}^+] = \sqrt{K_a \cdot C} = \sqrt{10^{-7} \times 0.001} = 10^{-5} \text{ M}$$

$$\Rightarrow \text{pH} = 5.0$$



$$\frac{K_w}{K_a} = 2 \times 10^{-6}$$

$$2 \times 10^{-6} = \frac{x \cdot x}{(6 \times 10^{-5} - x)} \Rightarrow x = 1 \times 10^{-5}$$

$$\therefore \text{pOH} = 5 \Rightarrow \text{pH} = 9$$

10. Ans. (B)

Sol. (P) $\text{Na}^+ > \text{Mg}^{2+} > \text{Al}^{3+}$ (Size of ion)

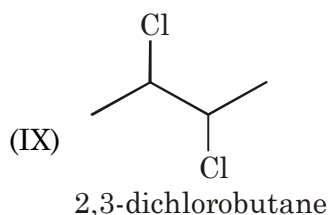
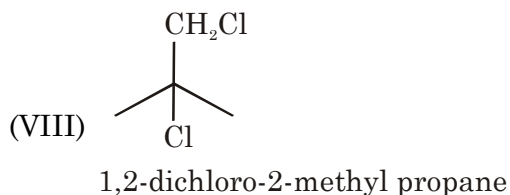
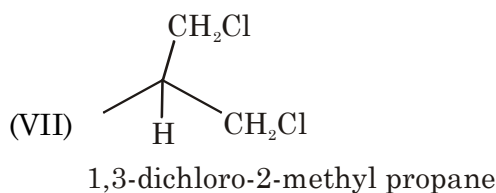
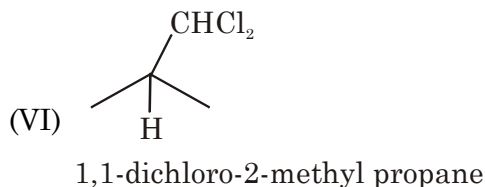
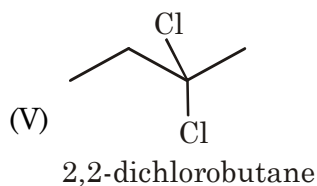
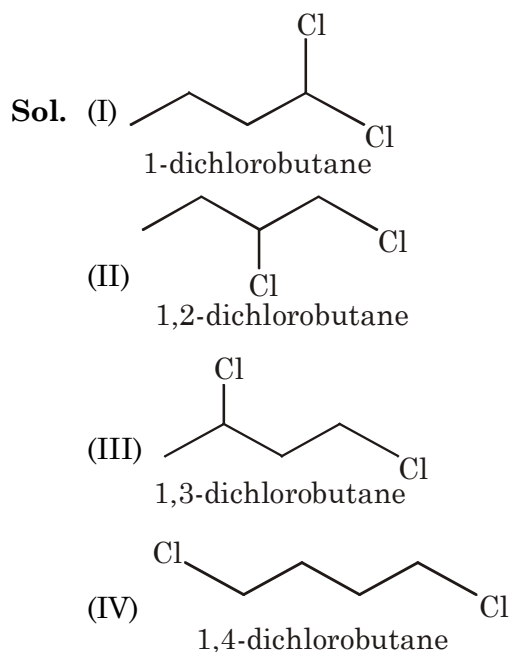
(Q) $\text{F}^- > \text{O}^-$ (First ionisation enthalpy)

(R) $\text{F} > \text{O} > \text{N}$ (Electronegativity)

(S) $\text{F}_2 < \text{O}_2 < \text{N}_2$ (Bond dissociation enthalpy)

SECTION-II

1. Ans.(9.00)



2. Ans.(5.00)

Sol. (i), (ii), (iii), (iv), (ix)

3. Ans.(4.00)

Sol. (1), (2), (4), and (5) statements are correct.

4. Ans.(7.00)

Sol. (a), (c), (d), (f), (g), (h), (j)

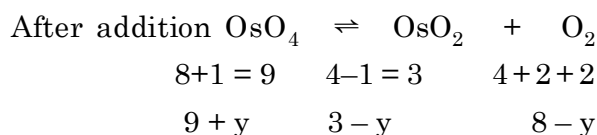
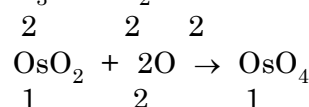
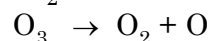
5. Ans.(2.50 to 2.60)

Sol. $\text{OsO}_4(\text{g}) \rightleftharpoons \text{OsO}_2(\text{g}) + \text{O}_2(\text{g})$

$$\begin{array}{rcl} 12 & & \\ 12-x & x & x \\ \frac{x^2}{12-x} = 2 & & x^2 = 24 - 2x \\ & & x^2 + 2x - 24 = 0 \\ & & x = +4 \end{array}$$

$$40 = \frac{n_{\text{O}_2} \times 32 + (4 - n_{\text{O}_2}) \times 48}{4}$$

$$n_{\text{O}_2} = 2 \text{ mole} = n_{\text{O}_3}$$



$$Q = \frac{8 \times 3}{9} = 2.6 > K \Rightarrow Q > K$$

Concentration of $\text{OsO}_2 < 3$

$$2 = \frac{(3-y)(8-y)}{(9+y)}$$

$$y \approx 0.5$$

Ans. 2.52 mol/L

6. Ans.(3.60)

$$\text{Sol. } -740 = 150 + \frac{3}{2} \times 200 + 350 + q - 1000$$

$$q \times 3 \times \Delta H_{\text{EG}} = -540;$$

$$q_1 = \frac{540}{3} = 180, \frac{q_1}{50} = 3.6$$

7. Ans. (2.00)

Sol. $\text{CH}_3\text{—CHBr—CH}_2\text{CH}_3$ is the only chiral molecule.

8. Ans. (2.00)

$$\begin{array}{l} \text{Sol. } -\frac{d[A]}{dt} = \frac{1}{x} \frac{d[B]}{dt} = \frac{1}{y} \frac{d[C]}{dt} \\ -\frac{(2-3)}{t} = \frac{1}{x} \frac{(2-0)}{t} \Rightarrow x = 2 \\ -\frac{(2-3)}{t} = \frac{1}{y} \frac{(1-0)}{t} \Rightarrow y = 1 \end{array}$$

$$[A]_{\text{eq}} = 2, [B]_{\text{eq}} = 2, [C]_{\text{eq}} = 1$$

PART-3 : MATHEMATICS

SOLUTION

SECTION-I

1. **Ans. (A,D)**

Sol. $\left(\cos^2 x + \frac{1}{\cos^2 x}\right)(1 + \tan^2 2y)(3 + \sin 3z) = 4$

$$\therefore \cos^2 x + \frac{1}{\cos^2 x} \geq 2,$$

$$1 + \tan^2 2y \geq 1, 2 \leq 3 + \sin 3z \leq 4$$

So, the only possibility is

$$\cos^2 x + \frac{1}{\cos^2 x} = 2, 1 + \tan^2 2y = 1, 3 + \sin 3z = 2$$

$$\Rightarrow \cos x = \pm 1$$

$$\Rightarrow x = m\pi$$

$$\tan 2y = 0$$

$$\Rightarrow y = \frac{n\pi}{2}$$

$$\sin 3z = -1$$

$$\Rightarrow z = (4p-1)\frac{\pi}{6}; m, n, p \in \mathbb{I}$$

2. **Ans. (A,C)**

Sol. Let $g(x) = ax^2 + b; a \neq 0$

$$L_1 = \lim_{x \rightarrow 0} f(x) = \lim_{x \rightarrow 0} \left(\frac{a}{x^2} + b \right) e^{-\frac{1}{x^2}}$$

Put $x = \frac{1}{t}$

$$L_1 = \lim_{t \rightarrow \infty} \frac{at^2 + b}{e^{t^2}} = 0$$

$$\text{Also, } f'(0) = \lim_{h \rightarrow 0} \frac{g\left(\frac{1}{h}\right) \cdot e^{-\frac{1}{h^2}} - 0}{h}$$

$$h = \frac{1}{t}$$

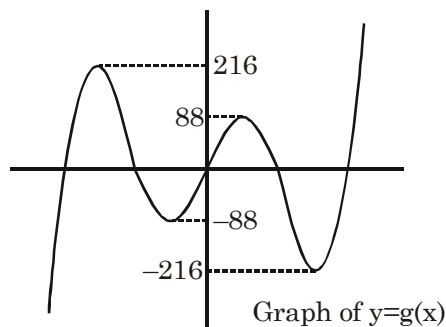
$$\Rightarrow f'(0) = \lim_{t \rightarrow \infty} g(t) \cdot e^{-t^2} \cdot t$$

$$= \lim_{t \rightarrow \infty} \frac{g(t) \cdot t}{e^{t^2}} = 0$$

$\Rightarrow f(x)$ is continuous and differentiable at $x = 0$

3. **Ans. (A,B,C)**

Sol. Let $g(x) = 3x^5 - 50x^3 + 135x$



$$g'(x) = 0 \text{ for } x = \pm 1, \pm 3$$

$$g(1) = 88; g(-1) = -88; g(3) = -216; g(-3) = 216$$

Now consider $g(x) = a$

4. **Ans. (A,C)**

Sol. $\tan \frac{A+B}{2} = \frac{\frac{5}{6} + \frac{20}{37}}{1 - \frac{5}{6} \cdot \frac{20}{37}} = \frac{305}{122}$

$$\Rightarrow \tan \frac{C}{2} = \frac{122}{305}$$

$$\tan \frac{B}{2} > \tan \frac{C}{2} \Rightarrow \angle B > \angle C$$

$$\Rightarrow \frac{\Delta}{s(s-b)} > \frac{\Delta}{s(s-c)}$$

$$\Rightarrow s-b < s-c$$

$$\Rightarrow b > c$$

Similarly we can get $a > b$

5. **Ans. (A,C,D)**

Sol. For A, D, assume $g(x) = x^n f(x)$

For C; apply LMVT in $[0, 4]$ and $[4, 5]$

6. **Ans. (A,C)**

Sol. Differentiating $fg = 1$, we get $f'g + g'f = 0$

$$\Rightarrow \frac{f'}{f} = -\frac{g'}{g}$$

$$\text{Differentiating again, we get } f''g + 2f'g' + g''f = 0$$

$$\Rightarrow \frac{f''}{f'} \cdot \frac{g}{g'} + 2 + \frac{g''}{g'} \cdot \frac{f}{f'} = 0$$

$$\Rightarrow \frac{f''}{f'} \cdot \frac{g}{g'} + 2 - \frac{g''}{g'} \cdot \frac{g}{g'} = 0 \Rightarrow \frac{g}{g'} \left(\frac{f''}{f'} - \frac{g''}{g'} \right) = -2$$

$$\Rightarrow \left(\frac{f''}{f'} - \frac{g''}{g'} \right) = \frac{-2g'}{g} \text{ or } \frac{2f'}{f}$$

7. Ans. (C)

Sol. (P) Draw the graph

If $|x| \leq 2\pi \Rightarrow$ both graphs cuts at 4 points

$$\therefore m = 4$$

If $|x| \leq 3\pi \Rightarrow$ both graphs cuts at $\therefore n = 6$

(Q) Draw the graphs

$$\Rightarrow m = 2$$

$$n = 3$$

$$\therefore m + n = 5,$$

$$n - m = 1$$

(R) $[\sin x] = \cos x$

$$-1 \leq \sin x \leq 1$$

$$\therefore \cos x = -1, 0, 1 \Rightarrow m = 0, n = 0$$

(S) make quad in $\sin x \Rightarrow n = 2, m = 0$

8. Ans. (A)

$$\text{Use } p_1 = \frac{2\Delta}{a}, p_2 = \frac{2\Delta}{b}, p_3 = \frac{2\Delta}{c}$$

$$(P) \frac{3}{\frac{1}{p_1} + \frac{1}{p_2} + \frac{1}{p_3}} \leq \sqrt[3]{p_1 p_2 p_3} \quad (\text{HM} \leq \text{GM})$$

$$(Q) \frac{\cos A}{p_1} + \frac{\cos B}{p_2} + \frac{\cos C}{p_3} = \frac{a \cos A + b \cos B + c \cos C}{2\Delta} = \frac{R(\sin 2A + \sin 2B + \sin 2C)}{2\Delta} \quad (\because a = 2R \sin A)$$

$$= \frac{R \cdot 4 \cdot \sin A \cdot \sin B \cdot \sin C}{2\Delta} = \frac{4R}{2\Delta} \cdot \frac{abc}{8R^3} = \frac{1}{R}$$

$$(R) \frac{b^2}{c} \cdot \frac{2\Delta}{a} + \frac{c^2}{a} \cdot \frac{2\Delta}{b} + \frac{a^2}{b} \cdot \frac{2\Delta}{c} = 2\Delta \left(\frac{a^3 + b^3 + c^3}{abc} \right)$$

$$\text{Now, } \frac{a^3 + b^3 + c^3}{3} \geq abc \quad (\text{AM} \geq \text{GM})$$

$$\frac{a^3 + b^3 + c^3}{abc} \geq 3 \Rightarrow 2\Delta \cdot \left(\frac{a^3 + b^3 + c^3}{abc} \right) \geq 6\Delta.$$

$$(S) \Sigma p_1^{-2} = \frac{\Sigma a^2}{4\Delta^2}$$

9. Ans. (D)

Sol. (P) $f''(x) > 0 \forall x \in \mathbb{R} \Rightarrow f'(x)$ is an increasing function.

$$\text{Now } g'(x) = -f'(4-x) + f'(2+x)$$

$$\text{If } g'(x) > 0 \Rightarrow f'(2+x) > f'(4-x)$$

$$\Rightarrow 2+x > 4-x \text{ or } x > 1$$

$$(Q) f'(x) = 3(x-1)(x+1) \Rightarrow f'(x) = 0$$

has roots $x = -1, 1$

$f(x) = 0$ will have exactly one real root if $f(-1)f(1) > 0$

$$\Rightarrow (a+2)(a-2) > 0 \Rightarrow a < -2 \text{ or } a > 2$$

$$(R) f'(x) = -\sin x + a^2 \geq 0 \forall x \in \mathbb{R}$$

$$\Rightarrow a^2 \geq \sin x \forall x \in \mathbb{R}$$

$$(S) f'(x) = 2e^x + ae^{-x} + 2a + 1 = 2e^{-x}(e^x + a)\left(e^x + \frac{1}{2}\right)$$

$f(x)$ increases for all x if $f'(x) \geq 0 \forall x \in \mathbb{R}$

10. Ans. (A)

$$\text{Sol. (P) } f'(x) = 12(x-1)(x+1)(x-2)$$

$\therefore$ Critical points are $x = -1, 1, 2$

$$f(-1) = -18$$

$$f(2) = 14$$

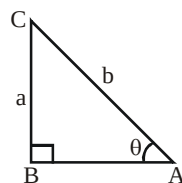
$$f(3) = 46$$

$$\Rightarrow m = -18$$

$$M = 46$$

(Q) $x = 0$ and 3 are repeated roots of $f(x) = 0$, hence $f(x)$ is derivable at all points

$$(R) A = \frac{1}{2}a\sqrt{b^2 - a^2} = \frac{1}{2}a\sqrt{(4-a)^2 - a^2}$$



$$= \frac{1}{2}a\sqrt{16-8a}$$

$$\frac{dA}{da} = \frac{1}{2} \left[a \left(\frac{-8}{2\sqrt{16-8a}} \right) + \sqrt{16-8a} \right] = 0$$

$$-4a + 16 - 8a = 0$$

$$\Rightarrow 16 - 12a = 0 \Rightarrow a = \frac{4}{3} \text{ (point of maxima)}$$

$$b = 4 - \frac{4}{3} = \frac{8}{3}$$

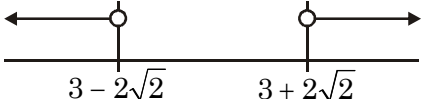
$$\sin \theta = \frac{a}{b} = \frac{4/3}{8/3} = \frac{1}{2} \Rightarrow \theta = \frac{\pi}{6}$$

$$\begin{aligned}
 \text{(S)} \quad y &= x^{(\sin x)^y} \\
 \Rightarrow \ell ny &= (\sin x)^y \ell nx \\
 \Rightarrow \ell ny &= e^{y \ell n(\sin x)} \cdot \ell nx \\
 \Rightarrow \frac{1}{y} \cdot \frac{dy}{dx} &= \frac{1}{x} e^{y \ell n(\sin x)} + \ell nx \cdot e^{y \ell n(\sin x)} \\
 &\quad \times \left(y \cdot \frac{1}{\sin x} \cdot \cos x + y \cdot \ell n(\sin x) \right) \\
 \Rightarrow \frac{1}{y} \frac{dy}{dx} &= \frac{1}{x/2} + 0 \Rightarrow \frac{dy}{dx} = 1 \\
 &\quad \left(\text{since at } x = \frac{\pi}{2}, y = \frac{\pi}{2} \right)
 \end{aligned}$$

SECTION-II

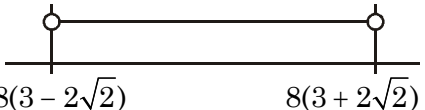
1. Ans. 2007.00

Sol. $f(x) = x^3 - 2kx^3 - 4kx + k^2 = 0$
 note that $f(0) = k^2 > 0$
 $f(1) > 0$
 $\Rightarrow 1 - 2k - 4k + k^2 > 0$
 $\Rightarrow k^2 - 6k + 1 > 0$
 $\Rightarrow [k - (3 + 2\sqrt{2})][k - (3 - 2\sqrt{2})] > 0$



....(1)

Also $f(4) < 0$
 $\Rightarrow 64 - 32k - 16k + k^2 < 0$
 $\Rightarrow k^2 - 48k + 64 < 0$
 $\Rightarrow (k - 24)^2 < 512$
 $\Rightarrow (k - 24 + 16\sqrt{2})(k - 24 - 16\sqrt{2}) < 0$
 $\Rightarrow [k - 8(3 - 2\sqrt{2})][k - 8(3 + 2\sqrt{2})] < 0$



....(2)

$(1) \cap (2) \Rightarrow 3 + 2\sqrt{2} < k < 8(3 + 2\sqrt{2})$
 $3 + \sqrt{8} < k < 8(3 + \sqrt{8})$
 $\therefore a = 3; b = 8$
 $(a + b)^3 + (ab + 2)^2$
 $\Rightarrow (11)^3 + (26)^2 = 1331 + 676 = 2007$

2. Ans. 2.00

Sol. $\lim_{x \rightarrow \infty} \frac{\log(x^2 + e^x)}{\log(x^4 + e^{2x})} = \lim_{x \rightarrow \infty} \frac{\log e^x \left(1 + \frac{x^2}{e^x} \right)}{\log e^{2x} \left(1 + \frac{x^4}{e^{2x}} \right)}$

$$\Rightarrow \lim_{x \rightarrow \infty} \frac{x + \log \left(1 + \frac{x^2}{e^x} \right)}{2x + \log \left(1 + \frac{x^4}{e^{2x}} \right)} = \frac{1 + \frac{1}{x} \log \left(1 + \frac{x^2}{e^x} \right)}{2 + \frac{1}{x} \log \left(1 + \frac{x^4}{e^{2x}} \right)}$$

$$\Rightarrow \ell = \frac{1}{2}$$

Similarly we can get $m = \frac{1}{2}$

$$\therefore \left[\frac{2\ell}{m} \right] = \left[\frac{2 \times \frac{1}{2}}{\frac{1}{2}} \right] = 2$$

3. Ans. 3.00

Sol. $4 \leq \text{LHS} \leq 16$
 $2 \leq \text{RHS} \leq 4$
 hence equality can occur at 4, which is possible if $x = \pi, 3\pi, 5\pi$
 $\Rightarrow 3$ solutions

4. Ans. 8.00

Sol. Let O be the point of intersection of the medians of triangle ABC. Then the area of ΔABC is three times of that ΔAOC . Now, in

ΔAOC , $AO = \frac{10}{3}$ Therefore, applying the

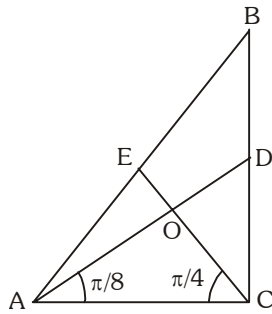
sine rule to ΔAOC ,

we get $\frac{OC}{\sin\left(\frac{\pi}{8}\right)} = \frac{AO}{\sin\left(\frac{\pi}{4}\right)}$

$$\Rightarrow OC = \frac{10}{3} \cdot \frac{\sin\left(\frac{\pi}{8}\right)}{\sin\left(\frac{\pi}{4}\right)}$$

area of $\Delta AOC = \frac{1}{2} AO \cdot OC \cdot \sin \angle AOC$

$$= \frac{1}{2} \cdot \frac{10}{3} \cdot \frac{10}{3} \cdot \frac{\sin\left(\frac{\pi}{8}\right)}{\sin\left(\frac{\pi}{4}\right)} \cdot \sin\left(\frac{\pi}{2} - \frac{\pi}{8}\right)$$



$$= \frac{50}{9} \cdot \frac{\sin\left(\frac{\pi}{8}\right) \cos\left(\frac{\pi}{8}\right)}{\sin\left(\frac{\pi}{4}\right)} = \frac{50}{18} = \frac{25}{9}$$

$$\therefore \text{area of } \triangle ABC = 3 \cdot \frac{25}{9} = \frac{25}{3} = \frac{p^2}{q} \Rightarrow p = 5, q = 3$$

$$\therefore p + q = 8$$

5. Ans. 1024.00

$$\text{Sol. } f(x) = \frac{3(x^2 + 1) + ax + b - 3}{1 + x^2}; f(x) = 3 + \frac{ax + b - 3}{1 + x^2}$$

$$y = 3 + \frac{ax + b - 3}{1 + x^2}$$

for y to lie in $[-4, 3)$

$$ax + b - 3 < 0 \quad \forall x \in \mathbb{R}$$

this is possible only if $m = 0$

$$\text{when, } a = 0 \text{ then } y = 3 + \frac{b - 3}{1 + x^2}$$

note that $b - 3 = 0$ then $b < 3$

$$\text{if } x \rightarrow \infty, y_{\max} \rightarrow 3^-$$

now $y_{\min}$ occurs at $x = 0$ (as $1 + x^2$ is minimum)

$$y_{\min} = 3 + b - 3 = b \Rightarrow b = -4$$

6. Ans. 0.00

$$\text{Sol. } \therefore f(x - y), f(x) \cdot f(y), f(x + y) \text{ are in AP}$$

$$\therefore 2f(x) \cdot f(y) = f(x - y) + f(x + y) \quad \dots(i)$$

Replacing x by y and y by x, then

$$2f(y) \cdot f(x) = f(y - x) + f(y + x) \quad \dots(ii)$$

From equations (i) and (ii), we get

$$f(x - y) = f(y - x)$$

Put $y = 0$, then

$$f(x) = f(-x) \Rightarrow f(2) = f(-2)$$

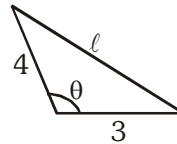
Differentiating w.r.t. x, then

$$f'(x) = -f'(-x)$$

$$\Rightarrow f'(x) + f'(-x) = 0 \Rightarrow f'(2) + f'(-2) = 0$$

7. Ans. 7.00

Sol.



Let angle between the hands be θ and distance between tips be ℓ

$$\ell = \sqrt{25 - 24 \cos \theta}$$

we have to find ℓ when $\frac{d\ell}{d\theta}$ is maximum

$$\frac{d\ell}{d\theta} = \frac{12 \sin \theta}{\sqrt{25 - 24 \cos \theta}} \text{ to maximum } \frac{d\ell}{d\theta}$$

$$\frac{d^2\ell}{d\theta^2} = 12 \left\{ \frac{\sqrt{25 - 24 \cos \theta} \cdot \cos \theta - \frac{24 \sin^2 \theta}{2\sqrt{25 - 24 \cos \theta}}}{(25 - 24 \cos \theta)} \right\} = 0$$

$$(25 - 24 \cos \theta) \cos \theta - 12 \sin^2 \theta = 0$$

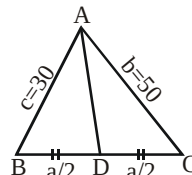
$$12 \cos^2 \theta - 25 \cos \theta + 12 = 0$$

$$\cos \theta = \frac{3}{4} \quad \cos \theta = \frac{4}{3} \text{ (Not possible)}$$

$$\text{hence } \ell = \sqrt{25 - 24 \cdot \frac{3}{4}} = \sqrt{7}$$

8. Ans. 5.00

$$\text{Sol. } \cos A = \frac{1}{2} = \frac{2500 + 900 - a^2}{2 \cdot 50 \cdot 30}$$



$$\Rightarrow 1500 = 2500 + 900 - a^2$$

$$\Rightarrow a^2 = 1900$$

$$\Rightarrow a = \sqrt{19} \cdot 10$$

$$\text{Also, } \cos B = \frac{1900 + 900 - 2500}{2 \cdot 30 \cdot \sqrt{19} \cdot 10}$$

$$\Rightarrow \cos B = \frac{300}{2 \cdot 30 \cdot \sqrt{19} \cdot 10} = \frac{1}{2\sqrt{19}}$$

$$\text{Also, } \cos B = \frac{c^2 + \frac{a^2}{4} - AD^2}{2 \cdot c \cdot \frac{a}{2}}$$

$$AD^2 = c^2 + \frac{a^2}{4} - ac \cos B$$

$$\Rightarrow AD^2 = 900 + \frac{1900}{4} - \sqrt{19} \cdot 10 \cdot 30 \cdot \frac{1}{2\sqrt{19}}$$

$$\Rightarrow AD^2 = 900 + 475 - 150 = 1225$$



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2020 - 2021)

JEE(Main)

PART TEST

09-01-2021

JEE(Main + Advanced) : ENTHUSIAST COURSE (SCORE-I)

PAPER-2

PART-1 : PHYSICS

SOLUTION

SECTION-I

1. Ans. (C)

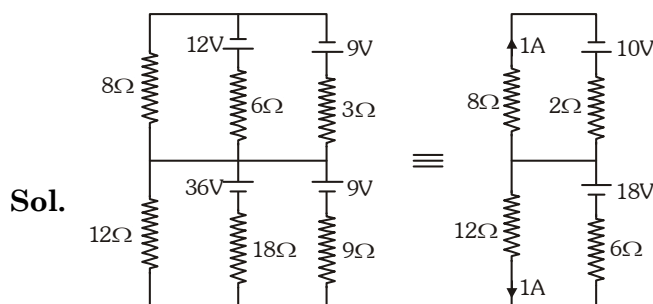
Sol. $R_{eq} = f(1-f)R_0$

R_{eq} is more for $f = 1/2$
 $\Rightarrow$ reading is min.

2. Ans. (A)

Sol. $R_{Cr} = \frac{12(4)}{12+4} = 3\Omega$

3. Ans. (A)



Hence, current through 8Ω is 1 A and

$$V_B - V_D = (V_B - V_E) + (V_E - V_D) \\ = -8 + 12 = 4V$$

4. Ans. (B)

Sol. $I = \frac{48}{6} + \frac{48}{24} + \frac{48}{24} = 12A$

It will divide in 6A, 4A & 2A in 2nd parallel segment further (A) have reading 4A.

5. Ans. (A)

Sol. All non-zero digits and trailing zero after decimal are significant.

6. Ans. (D)

Sol. Conceptual.

7. Ans. (A)

Sol. Error = $\frac{l.c.}{N}$

8. Ans. (A)

Sol. Experimental.

9. Ans. (B)

Sol. More focal length of objective will cause more tube length of the microscope.

10. Ans. (D)

Sol. Initially capacitor is closed switch, @ later time it is open.

11. Ans. (A)

Sol. $\omega_A = -(K-1)\frac{CV^2}{2}$

$$\omega_B = -\left(1 - \frac{1}{K}\right)\frac{CV^2}{2}$$

12. Ans. (D)

Sol. Inner surface will have Q & -Q

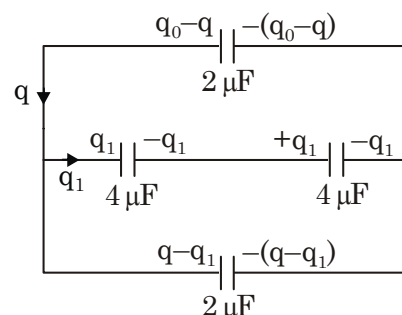
$$\Rightarrow \text{Heat} = \frac{Q^2 d}{2A \epsilon_0}$$

13. Ans. (A)

Sol. Force between the plates depends on the charges on plate. Which are constant in case of isolated capacitor.

14. Ans. (C)

Sol. The final charges on various capacitors is as shown in figure.



Using KVL

$$\frac{q_0 - q}{2} - \frac{q_1}{4} - \frac{q_1}{4} = 0 \Rightarrow q_0 = q + q_1$$

$$\frac{q_0 - q}{2} - \frac{q - q_1}{2} = 0 \Rightarrow q_0 = 2q - q_1$$

Solving above equations we get,

$$q_1 = \frac{q_0}{3}$$

Charge on C_4 is $q - q_1 = \frac{q_0}{3}$

15. Ans. (B)

Sol. We have

$$\eta = \frac{2r^2 (\sigma - \rho)g}{9 V_r}$$

$$\Rightarrow \eta = \frac{2 (0.01)^2 \times 1.75 \times 10^3 \times 9.8}{3.5 \times 10^{-3}}$$

$$\therefore \eta = 109 \text{ N s}^{-1} \text{ m}^{-2}$$

16. Ans. (C)

Sol. Mass of water in first tube is

$$m = \pi r^2 h \rho$$

Surface tension

$$\sigma = \frac{h \rho g r}{2} = \frac{h' \rho g r'}{2}$$

h' = height to which water rises

r' = radius, $r' = 2r$; $h' = \frac{h}{2}$

$$m' = \pi r'^2 h'$$

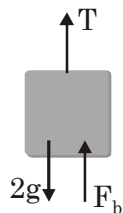
$$\rho = \pi (2r)^2 \frac{h}{2} \rho = \pi r^2 \rho$$

$$\rho = 2m = 2 \times 5 = 10 \text{ g}$$

17. Ans. (C)

Sol. Buoyancy force,

$$F_b = V_{\rho g} = \frac{2}{10 \times 10^3} \times 10^3 \times 10 = 2 \text{ N}$$



Free body diagram of block is as shown:

$$\Rightarrow T + F_b = 20 \Rightarrow T = 18 \text{ N}$$

The reading of S_1 is equal to tension in string.

The effect of putting the mass inside the water is equivalent to adding an amount of water equal to volume of block.

So, total force on S_1 is $5g + 2 = 52 \text{ N}$.

18. Ans. (C)

Sol. Velocity of efflux (v) = $\sqrt{2gd}$. Time taken

$$(t) = \sqrt{\frac{2h}{g}}$$

$\therefore$ Range R = velocity $\times$ time

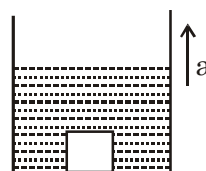
$$= vt = \sqrt{2gd} \times \sqrt{\frac{2h}{g}} = \sqrt{4dh}$$

$$R^2 = 4dh \text{ or } d = \frac{R^2}{4h}$$

19. Ans. (A)

Sol. $B_i = \rho_w v_g$

$$N_i = mg - \rho_w v_g$$



$$g_{\text{eff}} = g + a$$

$$B_f = \rho_w v(g + a)$$

$$N_f = m(g + a) - \rho_w v(g + a)$$

$$= N_i \frac{(g + a)}{g}$$

20. Ans. (A)

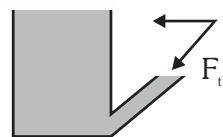
Sol. Let v is the speed of liquid which comes out from nozzle, then by applying Bernoulli's theorem, we get

$$\frac{\rho v^2}{2} + \frac{\rho g H}{4} + p_0 = p_0 + \rho g H$$

$$\Rightarrow \frac{\rho v^2}{2} = \frac{3\rho g H}{4} \Rightarrow v = \sqrt{2g \times \frac{3H}{4}}$$

Thrust force, $F = \rho a v^2$

$F_{\text{th}} \cos 45^\circ = kx$, where x is compression in spring.



Solving above equation we get,

$$x = \frac{3\rho a g H}{2\sqrt{2}k}$$

SECTION-II

1. Ans. 3.00

Sol. Pressure at bottom of vessel,

$$p = \rho g \times 6 \text{ cm}$$

$$= 1000 \times 10 \times \frac{6}{100}$$

Force acting at base,

$$F = pA_2 = 600 \times 50 \times 10^{-4} = 3 \text{ N}$$

2. Ans. 51.24

Sol. $5.10 + \frac{24}{50}(0.05) = 5.124 \text{ cm}$

3. Ans. 6.00

Sol. Energy initial = $\frac{q^2}{2C}$

Energy final = $\frac{q^2}{3C}$

4. Ans. 1.00

Sol. $\frac{10R}{R+1} = \eta(0.3)$

$\frac{5R}{R+2} = \eta(0.1)$

Solving $R = 1\Omega$

5. Ans. 0.50

Sol. Let p_1 and p_2 be air pressure on upper and lower surfaces of wing. From Bernoulli's theorem.

$$p_1 + \frac{1}{2}\rho v_1^2 = p_2 + \frac{1}{2}\rho v_2^2$$

where $v_1 = \sqrt{2}v$ Let $v_2 = v$

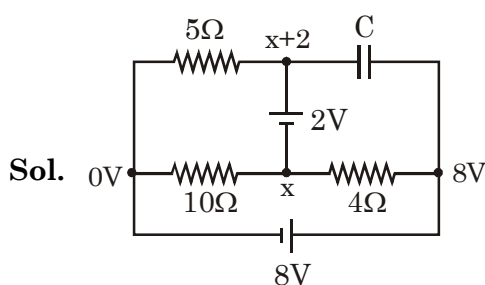
$\therefore$ Pressure difference,

$$p_2 - p_1 = \frac{1}{2}\rho(v_1^2 - v_2^2)$$

$$= \frac{\rho}{2}(2v^2 - v^2) = \frac{\rho v^2}{2}$$

$\therefore$ Force of dynamic lift = $\frac{1}{2}\rho v^2 A_1$

6. Ans. 3.09



Sol.

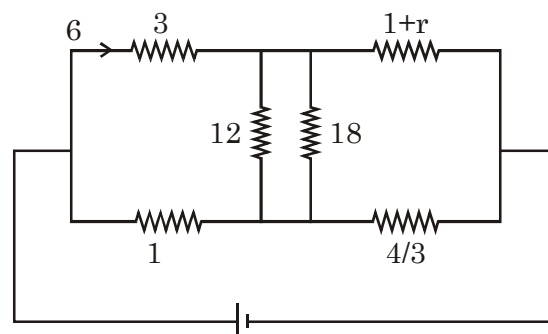
Net current at point $x = 0$.

$$\therefore \frac{0-x}{10} + \frac{8-x}{4} + \frac{0-(x+2)}{5} = 0$$

Potential across capacitor $\Rightarrow [(x+2) - 8]$

7. Ans. 42.00

Sol. On simplifying circuit will look like

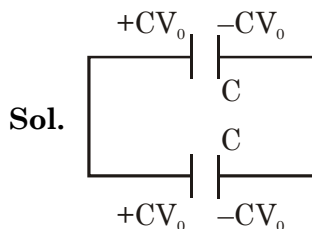


$\therefore$ Current through 12Ω is zero.

$$\Rightarrow \frac{3}{1} = \frac{1+r}{\left(\frac{4}{3}\right)}$$

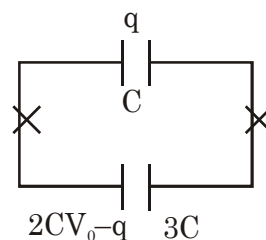
Get r and E

8. Ans. 6.00



Sol.

After



$$\frac{q}{C} = \frac{2CV_0 - q}{3C}$$

$$q = \frac{CV_0}{2}$$

$$\text{Potential difference} = \frac{q}{C}$$

9. Ans. 2.28

$$\text{Sol. } p = p_0 + h\rho g + \frac{2S}{r}$$

$$= 10^5 + 5 \times 10^4 + 2 \times \frac{0.07}{3.5 \times 10^{-6}} = 10^5 + 9 \times 10^4$$

$$p = \frac{\rho RT}{M}$$

$$\frac{p}{\rho} = \frac{p'}{\rho'}$$

$$\rho' = \frac{1.9 \times 10^5 \times 1.3}{10^5} = 2.28 \text{ kg/m}^3$$

10. Ans. 0.16

$$\text{Sol. } f = \frac{v}{2(\ell + 2e)} \text{ where } e = \text{end correction} = 0.6r$$

$$\therefore f = \frac{v}{2(\ell + 2 \times 0.6r)} = \frac{v}{2(\ell + 1.2r)}$$

$$\therefore \frac{\Delta f}{f} = \frac{\Delta v}{v} - \frac{\Delta(\ell + 1.2r)}{\ell + 1.2r}$$

$$= \frac{\Delta v}{v} - \frac{\Delta \ell + 1.2 \Delta r}{\ell + 1.2r}$$

$$\text{here } \frac{\Delta v}{v} = 0 \text{ (given)} \quad \frac{\Delta f}{f} \times 100$$

$$= - \frac{\Delta \ell + 1.2 \Delta r}{\ell + 1.2r} \times 100$$

for maximum % error : $\Delta \ell = 0.1$, $\Delta r = 0.05$

$$\left(\frac{\Delta f}{f} \times 100 \right)_{\max} = \frac{0.1 + 1.2 \times 0.05}{94 + 1.2 \times 5} \times 100$$

$$= 0.16\%$$

PART-2 : CHEMISTRY

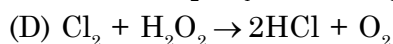
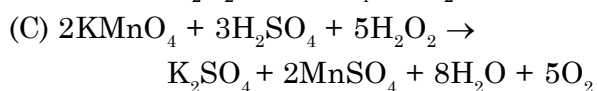
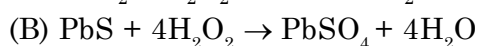
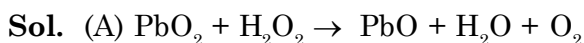
SOLUTION

SECTION-I

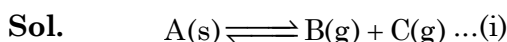
1. Ans. (D)

Sol. Nitrogen is the chiral centre.

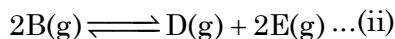
2. Ans. (B)



3. Ans. (C)



At eq. 0.05 0.1 0.2



0.1 0.05 0.1

$$P_{\text{Total}} = \frac{(0.1 + 0.2 + 0.05 + 0.1)RT}{V}$$

$$= \frac{0.45 \times 0.08 \times 300}{1} = 10.8 \text{ atm}$$

$$K_{C_1} = 0.1 \times 0.2 = 0.02$$

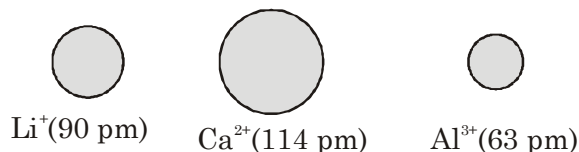
$$P_B = \frac{0.1}{0.45} \times 10.8 = 2.4 \text{ atm}$$

4. Ans. (A)

Sol. I and II are pair of tautomers.

5. Ans. (C)

Sol.



$$\text{LiF} = 1036 \text{ kJ/mole}$$

$$\text{CaF}_2 = 2630 \text{ kJ/mole}$$

$$\text{AlF}_3 = 5215 \text{ kJ/mole}$$

$$X > Z > Y$$

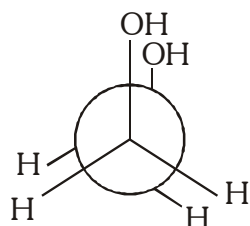
6. Ans. (C)

$$\text{Sol. } K_h = \frac{(1.6 \times 10^{-4})^2}{0.01} = 2.56 \times 10^{-6}$$

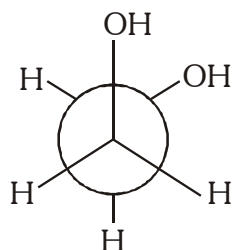
$$K_h = \frac{K_w}{K_b} \Rightarrow K_b = 3.9 \times 10^{-9}$$

7. Ans. (C)

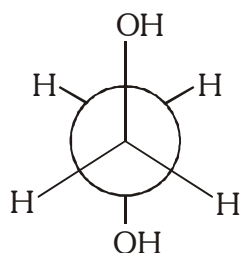
Sol. P is $\text{CH}_2 = \text{CH}_2$, Q is MnO_4^- and
R is $\text{CH}_2(\text{OH})-\text{CH}_2(\text{OH})$



(I)



(II)



(III)

Stability $\text{II} > \text{III} > \text{I}$

8. Ans. (C)

Sol. With increasing in electronegative difference ionisation occur readily.

9. Ans. (C)

Sol. $S = [\text{Ag}^+] + [\text{Ag}(\text{CN})_2^-]$

$$= \frac{K_{sp}}{[\text{CN}^-]} + K_f \cdot K_{sp} \cdot [\text{CN}^-]$$

For minimum solubility: $\frac{dS}{d[\text{CN}^-]} = 0$

$$\text{or, } -\frac{K_{sp}}{[\text{CN}^-]^2} + K_f \cdot K_{sp} = 0$$

$$\Rightarrow [\text{CN}^-] = \sqrt{\frac{1}{K_f}} = 2.58 \times 10^{-9} \text{ M}$$

10. Ans. (B)

Sol. *Gauche* and *anti*-conformers are stereoisomers not related by mirror image relationship, hence distereomer.

11. Ans. (D)

Sol. There is large difference in I.E_2 and I.E_3 hence it can have two electrons in valence shell.

12. Ans. (B)

Sol. $V_{\text{Diamond}} < V_{\text{Graphite}}$

On increasing pressure, equilibrium will shift toward diamond.

13. Ans. (D)

Sol. To exhibit G.I. compound need not to have π -bond.

14. Ans. (B)

Sol. 1 mL H_2O_2 of 30 volume liberates 30 mL O_2 at NTP.

$\therefore$ 100 mL of H_2O_2 of 30 volume liberate $30 \times 100 \text{ mL O}_2$ at NTP or 3L O_2 at NTP.

15. Ans. (C)

Sol. (ii) NaCN (0.5 moles) + HCN (0.5 moles)

(iii) NH_3 (1 mole) + NH_4Cl (1 mole)

(iv) CH_3COOH (1 mole) + NaOH (0.75 moles)

(v) NH_4Cl (1 mole) + NaOH (0.75 moles)

(vi) CH_3COONa (0.5 moles) + HCl (0.25 mole)

(vii) H_2CO_3 (1 mole) + NaOH (1.5 moles)

16. Ans. (B)

Sol. Specific rotation

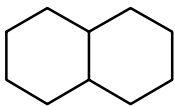
$$\frac{\text{observed rotation}}{l \times c} = \frac{30}{10 \times \frac{20}{200}} = 30^\circ$$

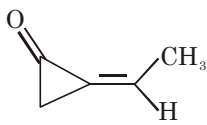
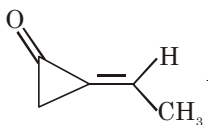
17. Ans. (A)

Sol. In second period E.A is poor. Thus $S > O$. Nitrogen having stable E.C. lower E.A. than O.

18. Ans. (C)

Sol. Yes, the above compound does show geometrical isomerism.

(X)  it forms cis & trans decalin.

(Y)  and  A
pair of geometrical isomers.

19. Ans. (B)



$$\frac{10 \times 0.1}{20} M \quad \frac{10 \times 0.1}{20} M \quad 0$$

Final 0 0 0.05 M

(10 ml HCl solution is needed for equivalent point)

At equivalent point $[H^+] = \sqrt{K_a \times C}$

$$= \sqrt{10^{-8} \times 0.05}$$

$$\therefore pH = -\log (5 \times 10^{-10})^{1/2} = 4.65$$

20. Ans. (D)

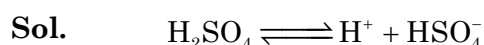
Sol.. (i) Z, (ii) Z, (iii) E, (iv) E.

SECTION-II

1. Ans. (130.00)

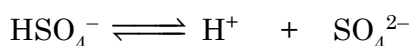
Sol. In general with increasing of coordination number, cationic size increases

2. Ans. (11.00)



$$CM \quad 0 \quad 0$$

Final ≈ 0 CM CM



Final $(C-x)M \quad (C+x)M \quad xM$

$$\text{Now, } 1.2 \times 10^{-2} = \frac{0.01 \times x}{C-x} \Rightarrow x = \frac{6}{11} C$$

$$\text{And } C+x=0.01 \Rightarrow C = \frac{0.01 \times 11}{17} M$$

3. Ans. (11.00)

Sol. (I) 4 (II) 3 (III) 4.

4. Ans. (5.00)

Sol. (a), (b), (d), (e), & (h) are incorrect.

5. Ans. (2.00)

Sol. (C) & (F).

Increase in temperature favours endothermic direction and increase in the pressure favours the direction of decreases in volume (moles of gases).

6. Ans. (10.00)

7. Ans. (5.00)

8. Ans. (8.00)

9. Ans. (4.00)

10. Ans. (8.00)

PART-3 : MATHEMATICS

SOLUTION

SECTION-I

1. Ans. (A)

Sol. $y = (1+x)(1+x^2)(1+x^4) \dots (1+x^{2^n})$

Multiplying numerator and denominator by $(1-x)$

$$\Rightarrow y = \frac{(1-x)(1+x)(1+x^2) \dots (1+x^{2^n})}{(1-x)}$$

$$\Rightarrow y = \left(\frac{1-x^{2^{n+1}}}{(1-x)} \right)$$

$$\therefore \frac{dy}{dx} = \frac{(1-x) \cdot \left\{ -2^{n+1} \cdot x^{2^{n+1}-1} \right\} - (1-x^{2^{n+1}})(-1)}{(1-x)^2}$$

$$\text{So } \left. \frac{dy}{dx} \right|_{x=0} = \frac{-2^{n+1} \cdot 0.1 + 1 - 0}{1^2} = 1$$

Hence (A) is correct answer.

2. Ans. (B)

Sol. $f'(x) = 8x + 24x^3 + 48x^5$

and

$$f''(x) = 8 + 72x^2 + 240x^4 \geq 0 \text{ for all } x \in \mathbb{R}$$

$$\therefore f'(x) = 0 \Rightarrow x(8 + 24x^2 + 48x^4) = 0$$

$$\Rightarrow x = 0. \text{ Since } f''(0) > 0$$

$\therefore f$ has a local minima at $x = 0$.

There is no other critical point and

$f'(x)$ exists

for all $x \in \mathbb{R}$

3. Ans. (D)

Sol. L.H.S.

$$= \sqrt{3} \sin \pi x + \cos \pi x = 2 \sin \left(\pi x + \frac{\pi}{6} \right) \leq 2$$

$$\text{and equality holds for } x = \frac{1}{3}$$

and R.H.S.

$$= x^2 - \frac{2}{3}x + \frac{19}{9} = \left(x - \frac{1}{3} \right)^2 + 2 \geq 2$$

equality holds if $x = \frac{1}{3}$.

Thus L.H.S. = R.H.S. for $x = \frac{1}{3}$ only.

Hence (D) is the correct answer.

4. **Ans. (C)**

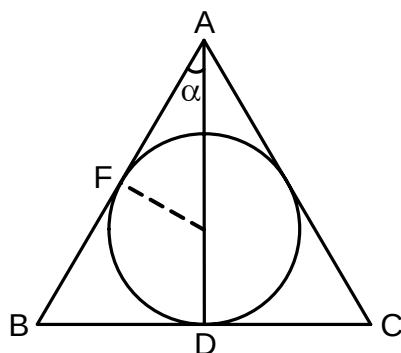
$$\begin{aligned} \text{Sol. } \sin A + \sin B &= \frac{c(a+b)}{c^2} = \frac{a+b}{c} \\ &= \frac{k \sin A + k \sin B}{k \sin C} \end{aligned}$$

Thus $C = \pi/2 \Rightarrow$ ABC is right angled.

Hence (C) is the correct answer.

5. **Ans. (C)**

$$\text{Sol. } 2s = AB + BC + CA = 2AB + 2BD$$



$$\Rightarrow s = AB + BD = AF + 2BD = r \cot \alpha + 2AD \tan \alpha$$

$$= r \cot \alpha + 2(r + r \operatorname{cosec} \alpha) \tan \alpha$$

$$= r(\cot \alpha + 2 \tan \alpha + 2 \sec \alpha)$$

$$\text{Find } \frac{ds}{d\alpha} \text{ and } \frac{ds}{d\alpha} = 0 \Rightarrow \alpha = \frac{\pi}{6}$$

$$\therefore 2s = 6r\sqrt{3}$$

6. **Ans. (D)**

$$\text{Sol. } f(x, y) = \sin 2x ;$$

$$\therefore \lim_{x \rightarrow 0} (1 + \tan x - x)^{\frac{1}{x^2(\sin 2x)}} = e^l$$

$$\text{where } l = \lim_{x \rightarrow 0} \frac{\tan x - x}{x^2 \sin 2x}$$

$$= \lim_{x \rightarrow 0} \frac{\tan x - x}{2x^3} = \frac{1}{6} ;$$

$$\text{Hence limit} = e^{\frac{1}{6}} \text{ Ans.}$$

7. **Ans. (B)**

$$\text{Sol. We have } \sin^4 x - 2 \cos^2 x + a^2 = 0$$

$$\Rightarrow y^2 - 2(1-y) + a^2 = 0$$

where $\sin^2 x = y$

$$\Rightarrow y^2 + 2y + a^2 - 2 = 0$$

$$\Rightarrow y = -1 \pm \sqrt{3 - a^2}$$

for y to be real.

$$\text{Discriminant} \geq 0$$

$$\Rightarrow 4 - 4(a^2 - 2) \geq 0$$

$$\Rightarrow a^2 \leq 3 \quad \dots(1)$$

But $\sin^2 x = y$, therefore $0 \leq y \leq 1$

$$\Rightarrow 0 \leq -1 + \sqrt{3 - a^2} \leq 1 \Rightarrow 1 \leq \sqrt{3 - a^2} \leq 2$$

$$\Rightarrow 1 \leq 3 - a^2 \leq 4$$

$$\Rightarrow 2 - a^2 \geq 0$$

$$\Rightarrow a^2 \leq 2 \quad \dots(2)$$

From (1) and (2),

$$a^2 \leq 2 \Rightarrow -\sqrt{2} \leq a \leq \sqrt{2}$$

Hence (B) is the correct answer.

8. **Ans. (C)**

Sol. Given that $A_0 A_1 A_2 A_3 A_4 A_5$ is a regular hexagon inscribed in a circle of radius 1.

$$\angle A_0 O A_1 = \frac{360^\circ}{6} = 60^\circ$$

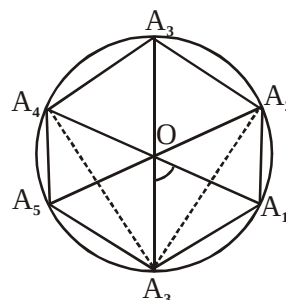
But in $\triangle O A_0 A_1$, $O A_0 = O A_1 = 1$

$$\therefore \angle O A_0 A_1 = \angle O A_1 A_0 = 60^\circ$$

$$\therefore \triangle O A_0 A_1 \text{ is an equilateral } \triangle$$

$$\therefore A_0 A_1 = 1 = A_1 A_2 = A_2 A_3$$

$$= A_3 A_4 = A_4 A_5 = A_5 A_0$$



$$\angle A_0 A_1 A_2 = 60^\circ + 60^\circ = 120^\circ$$

In $\triangle A_0 A_1 A_2$, using cosine law, we get

$$\cos 120^\circ = \frac{(A_0 A_1)^2 + (A_1 A_2)^2 - (A_0 A_2)^2}{2(A_0 A_1)(A_1 A_2)}$$

$$\Rightarrow -\frac{1}{2} = \frac{1+1-(A_0 A_2)^2}{2 \times 1 \times 1} \Rightarrow A_0 A_2 = \sqrt{3}$$

Thus by symmetry $A_0 A_4 = \sqrt{3}$

$$A_0 A_1 \cdot A_0 A_2 A_0 A_4 = 1 \cdot \sqrt{3} \cdot \sqrt{3} = 3$$

$\therefore$ (C) is the correct option.

9. Ans (B)

Sol. Given $f'(x) \leq \frac{1}{2}$

$$\Rightarrow \frac{d}{dx} \left(f(x) - \frac{x}{2} \right) \leq 0$$

$$\Rightarrow h(x) = f(x) - \frac{x}{2} \text{ is a decreasing that.}$$

Now $h(0) = 0$ and $h(x)$ is decreasing

$$\Rightarrow f(x) - \frac{x}{2} \leq 0$$

$$\Rightarrow f(x) \leq \frac{x}{2}$$

$$\Rightarrow f(x) \leq 1 \text{ since } x \in \{0, 2\}$$

10. Ans.(A)

Sol. $\frac{dy}{d\theta} = n \sec^{n-1} \theta \cdot \sec \theta \cdot \tan \theta - n \cdot \cos^{n-1} \theta \cdot (-\sin \theta)$

$$= n \left[\sec^n \theta \frac{\sin \theta}{\cos \theta} + \cos^{n-1} \theta \cdot \sin \theta \right]$$

$$= \frac{n \sin \theta}{\cos \theta} [\sec^n \theta + \cos^n \theta]$$

$$n \tan \theta (\sec^n \theta + \cos^n \theta)$$

$$\frac{dx}{d\theta} = \sec \theta \tan \theta + \sin \theta = \sec \theta \frac{\sin \theta}{\cos \theta} + \sin \theta$$

$$= \frac{\sin \theta}{\cos \theta} (\sec \theta + \cos \theta) = \tan \theta (\sec \theta + \cos \theta)$$

$$\therefore \frac{dy}{dx} = \frac{n \tan \theta (\sec^n \theta + \cos^n \theta)}{\tan \theta (\sec \theta + \cos \theta)}$$

$$= \frac{n (\sec^n \theta + \cos^n \theta)}{\sec \theta + \cos \theta}$$

$$\therefore \left(\frac{dy}{dx} \right)^2 = \frac{n^2 (\sec^n \theta + \cos^n \theta)^2}{(\sec \theta + \cos \theta)^2}$$

Hence (A) is correct answer.

11. Ans. (C)

Sol. $5 \cos^2 \theta - 3 \sin^2 \theta + 6 \sin \theta \cos \theta = 7$

$\Rightarrow$

$$5 \left(\frac{1 + \cos 2\theta}{2} \right) - 3 \left(\frac{1 - \cos 2\theta}{2} \right) + 3 \sin 2\theta = 7$$

$$\Rightarrow 4 \cos 2\theta + 3 \sin 2\theta = 6$$

$$\text{but } 4 \cos 2\theta + 3 \sin 2\theta \leq \sqrt{4^2 + 3^2} = 5$$

$\therefore$ Solution does not exist.

Hence (C) is the correct answer

12. Ans. (D)

Sol. Let $g(x) = f(x) - (ax^2 + bx + c)$.

Since, $g(x) = 0$ has 10 distinct real roots,

$\Rightarrow g'(x) = 0$ has minimum 9 distinct real roots,

$\Rightarrow g''(x) = 0$ has minimum 8 distinct real roots,

$\Rightarrow g'''(x) = 0$ has minimum 7 distinct real roots,

$\Rightarrow f'''(x) = 0$ has at least 7 distinct real roots.

13. Ans. (C)

Sol. $y^2 = p(x) \Rightarrow 2y y_1 = p'(x)$

$$\Rightarrow 2(y y_2 + y_1 y_1) = p''(x)$$

$$\Rightarrow y y_2 = \frac{1}{2} (p''(x) - 2y_1^2)$$

$$\Rightarrow y^3 y_2 = \frac{1}{2} (p''(x) y^2 - 2(y y_1)^2)$$

$$= \frac{1}{2} p''(x) \cdot p(x) - \left(\frac{1}{2} p'(x) \right)^2$$

$$\Rightarrow \frac{d}{dx} (y^3 y_2) = \frac{1}{2} [p'''(x) p(x) + p'(x) p''(x)] - 2 \times \left(\frac{1}{2} p'(x) \right) \times \left(\frac{1}{2} p''(x) \right)$$

$$\Rightarrow 2 \frac{d}{dx} (y^3 y_2) = p(x) p'''(x)$$

Hence (C) is correct answer.

14. Ans. (B)

Sol. Assume a function $h(x) = f(x) - 2g(x)$

$$h(0) = h(1) = 0$$

Using Rolle's theorem option B is true

15. Ans. (C)

Sol. Since A, C_1, G and B_1 are concyclic,

therefore $BG \cdot BB_1 = BC_1 \cdot BA$

$$\Rightarrow \frac{2}{3} (BB_1)^2 = \frac{c}{2} \cdot c$$

$$\Rightarrow \frac{2}{3} \cdot \frac{1}{4} (2a^2 + 2c^2 - b^2) = \frac{c^2}{2}$$

$$\Rightarrow 2a^2 + 2c^2 - b^2 = 3c^2$$

$$\Rightarrow b^2 + c^2 = 2a^2$$

Hence (C) is the correct answer.

16. Ans. (B)

Sol. $y^x = x^y \Rightarrow \log y^x = \log x^y$
 $\Rightarrow x \log y = y \log x$
 $\Rightarrow x \cdot \frac{1}{y} \frac{dy}{dx} + \log y \cdot 1 = y \cdot \frac{1}{x} + \log x \cdot \frac{dy}{dx}$
 $\Rightarrow \left(\frac{x}{y} - \log x \right) \frac{dy}{dx} = \frac{y}{x} - \log y$

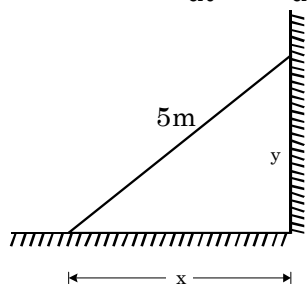
Hence (B) is correct ans

17. Ans. (C)

Sol. convert $\tan x$ into $\sec x$ and thus make quadratic eq in $\sec x$ and hence solve

18. Ans (A)

Sol. $x^2 + y^2 = 25 \Rightarrow 2x \frac{dx}{dt} + 2y \frac{dy}{dt} = 0$



but $\frac{dx}{dt} = 1.5 \text{ m/sec}$

$\therefore 2(x)(1.5) + 2(y) \frac{dy}{dt} = 0 \Rightarrow \frac{dy}{dt} = -\frac{1.5x}{y}$

when $x = 4, y = 3$

$\therefore \frac{dy}{dt} = -\frac{1.5 \times 4}{3} = -2 \text{ m/sec}$

$\therefore$ Height of the wall is decreasing at the rate of 2 m/sec.

19. Ans. (A)

Sol. $4x^3 + 4y^3 \frac{dy}{dx} = 0 \Rightarrow \frac{dy}{dx} = -\frac{x^3}{y^3}$

Equation of tangent,

$Y - y = -\frac{x^3}{y^3}(X - x)$

$\Rightarrow y^3 Y + x^3 X = x^4 + y^4 = a^4$

$\Rightarrow \frac{X}{a^4/x^3} + \frac{Y}{a^4/y^3} = 1$

Here, $P = \frac{a^4}{x^3}, q = \frac{a^4}{y^3}$

$\Rightarrow p^{-4/3} + q^{-4/3} = \frac{a^{-16/3}}{x^{-4}} + \frac{a^{-16/3}}{y^{-4}}$
 $= a^{-16/3} (x^4 + y^4) = a^{-16/3} (a^4) = a^{-4/3}$

20. Ans. (B)

Sol. $\therefore y = \sqrt{x} + \sqrt{y+y}$

$\Rightarrow (y^2 - x) = \sqrt{2y}$

or $(y^2 - x)^2 = 2y$

Differentiating both sides w.r.t. x , then

$2(y^2 - x) \left(2y \frac{dy}{dx} - 1 \right) = 2 \frac{dy}{dx}$

$\therefore \frac{dy}{dx} = \frac{(y^2 - x)}{2y^3 - 2xy - 1}$

SECTION-II

1. Ans. 25.00

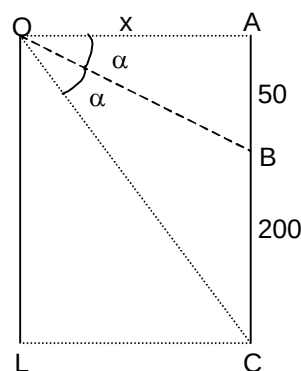
Sol. Let BC be the building and AB = 50 be the pole. Let O be the observer so that OA = x. If $\angle AOB = \alpha$ and $\angle AOC = 2\alpha$,

then $\tan 2\alpha = \frac{300}{x}$

$\Rightarrow \frac{300}{x} = \frac{2 \tan \alpha}{1 - \tan^2 \alpha} = \frac{\frac{100}{x}}{1 - \frac{2500}{x^2}} = \frac{100x}{x^2 - 2500}$

or $3x^2 - 7500 = x^2$ or $x^2 = 3750$

$\Rightarrow x = 25\sqrt{6}$



Hence

2. Ans. 7.00

Sol. $11 \sin x = x$

Draw graph

3. Ans. 1.00

Sol. $\left(\frac{dy}{dx} \right)_{(a,b)} = \frac{-a^2}{b^2}$

$a^3 + b^3 = c^3$ (1)

$a_1^3 + b_1^3 = c^3$ (2)

solving $\frac{a_1}{a} + \frac{b_1}{b} = -1$

4. Ans. 3.00

$$y = f(x) \Rightarrow x = f^{-1}(y) \Rightarrow x = g(y)$$

$$\text{Given } y = f(x) = \int_0^x \frac{dt}{\sqrt{1+t^3}}$$

$$\frac{dy}{dx} = \frac{1}{\sqrt{1+x^3}} \Rightarrow \frac{dx}{dy} = \sqrt{1+x^3}$$

$$g'(y) = \sqrt{1+g^3(y)}$$

$$g''(y) = \frac{3g^2(y)g'(y)}{2\sqrt{1+g^3(y)}}$$

$$\Rightarrow 2g''(y) = 3g^2(y) \frac{g'(y)}{\sqrt{1+g^3(y)}}$$

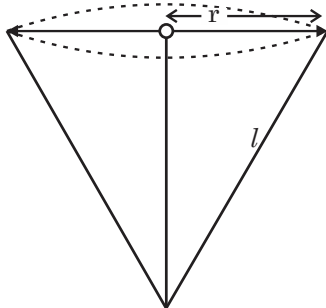
$$= 3g^2(y) \frac{\sqrt{1+g^3(y)}}{\sqrt{1+g^3(y)}} = 3g^2(y)$$

$$\Rightarrow 2g''(y) = 3g^2(y) \text{ Ans.}$$

5. Ans. 20.00

Sol. Here, $r = h$

$$\therefore l^2 = r^2 + h^2 = 2r^2$$



$$\text{or } h = \frac{l}{\sqrt{2}}, r = \frac{l}{\sqrt{2}}$$

$$V = \frac{1}{3}\pi r^2 h = \frac{1}{3}\pi \left(\frac{l}{\sqrt{2}}\right)^2 \left(\frac{l}{\sqrt{2}}\right) = \frac{\pi}{6\sqrt{2}} l^3$$

$$\therefore \frac{dV}{dt} = \frac{3\pi}{6\sqrt{2}} l^2 \frac{dl}{dt}$$

$$\Rightarrow 2 = \frac{3\pi}{6\sqrt{2}} (4)^2 \frac{dl}{dt}$$

$$\Rightarrow \frac{dl}{dt} = \frac{\sqrt{2}}{4\pi} \text{ cm/s}$$

$$\therefore \text{Here, } \lambda = \frac{\sqrt{2}}{4\pi}$$

$$\text{Then, } \frac{40\pi\sqrt{2}\lambda}{40\pi\sqrt{2} \times \frac{\sqrt{2}}{4\pi}} = 10 \times 2 = 20$$

6. Ans. 0.25

$$\text{Sol. } y = \tan^{-1} \sqrt{\frac{1+\cos x}{1-\cos x}}$$

$$= \tan^{-1} \left(\tan \left(\frac{\pi}{2} - \frac{x}{2} \right) \right) = \frac{\pi}{2} - \frac{x}{4}$$

$$\therefore \frac{dy}{dx} = -\frac{1}{4}$$

7. Ans. 0.25

Sol. $x = e^t \sin t$

$$y = e^t \cos t$$

$$\therefore e^t = \sqrt{x^2 + y^2} \text{ and } t = \tan^{-1} \frac{x}{y}$$

$$\therefore e^{\tan^{-1} \left(\frac{x}{y} \right)} = \sqrt{x^2 + y^2}$$

$$\tan^{-1} \left(\frac{x}{y} \right) = \frac{1}{2} \ln(x^2 + y^2)$$

differentiate both side :

$$\left(\frac{1}{1 + \frac{x^2}{y^2}} \right) \left(\frac{y \cdot 1 - x \cdot y'}{y^2} \right) = \frac{1}{2} \frac{(2x + 2yy')}{(x^2 + y^2)}$$

$$y' = \frac{y - x}{x + y}$$

$$\therefore y'' = \frac{(x + y)(y' - 1) - (y - x)(1 + y')}{(x + y)^2}$$

$$y''(x + y)^2 = 2(xy' - y)$$

8. Ans. 0.85 or 0.86

$$\text{Sol. } \therefore x = t^2 + 3t - 8 = 2 \text{ and } y = 2t^2 - 2t - 5 = -1$$

$$\therefore t = 2$$

$$\therefore \frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{4t - 2}{2t + 3} = \frac{6}{7} = 0.86$$

9. Ans. 0.25

$$\text{Sol. } f'(x) = \cos x - 2a \cos 2x - \cos 3x + 2a \geq 0 \quad \forall x \in \mathbb{R}$$

$$= \cos x - \cos 3x + 2a(1 - \cos 2x) \geq 0$$

$$= 2 \sin 2x \cdot \sin x + 4a \sin^2 x \geq 0$$

$$= 4 \sin^2 x (\cos x + a) \geq 0$$

$$\Rightarrow a \geq -\cos x$$

$$a \geq 1$$

$$\lambda = 2$$

10. Ans. 1.25

$$\text{Sol. } f(x) = x^3 - 6x^2 + ax + b$$

$$f'(x) = 3x^2 - 12x + a$$

$$f'(c) = 0 \Rightarrow f' \left(2 + \frac{1}{\sqrt{3}} \right) = 0 \Rightarrow a = 11$$